

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Rithika K	Reg. No.:	192421075
Section / Batch:	2024-2028	Date:	17/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Application: Weather Forecasting Simulation System

A weather simulation system uses two algorithms with complexities $O(n \log n)$ and $O(n^2 \log n)$. Analyze both algorithms using asymptotic notation and compare their growth rates. Explain how each algorithm performs as the dataset size increases. Determine which algorithm is more efficient for large-scale simulations. Justify your answer based on computational efficiency and scalability.

ANSWER:

Algorithm Used

The program simulates two algorithms with different time complexities:

- **Algorithm 1:** Sorting the weather data using the C++ STL `sort()` function, which has a time complexity of $O(n \log n)$.

- **Algorithm 2:** Sorting the data followed by nested loops that compare every pair of weather records, resulting in an overall complexity of **$O(n^2 \log n)$** .

Input

The program accepts:

- Number of weather records (**n**)
- Weather values such as temperature, humidity, or rainfall readings.

Output

The program displays:

- Sorted weather data (Algorithm 1)
- Total simulation value after pairwise computations (Algorithm 2)

Complexity Analysis

Algorithm 1

Sorting requires

$$T(n) = O(n \log n) \quad T(n) = O(n \log n) \quad T(n) = O(n \log n)$$

Since the sorting algorithm dominates the execution time,

$$O(n \log n) \quad \boxed{O(n \log n)} \quad O(n \log n)$$

Best Case

$$\Omega(n \log n) \quad \Omega(n \log n) \quad \Omega(n \log n)$$

Average Case

$$\Theta(n \log n) \quad \Theta(n \log n) \quad \Theta(n \log n)$$

Worst Case

$$O(n \log n) \quad O(n \log n) \quad O(n \log n)$$

Algorithm 2

Sorting requires

$$O(n \log n) \quad O(n \log n) \quad O(n \log n)$$

Nested loops require

$$O(n^2) \quad O(n^2) \quad O(n^2)$$

Therefore,

$$T(n) = O(n \log n) + O(n^2) \quad T(n) = O(n \log n) + O(n^2) \quad T(n) = O(n \log n) + O(n^2)$$

Since

$$n^2 > n \log n \quad n^2 > n \log n \quad n^2 > n \log n$$

the dominant term becomes

$$O(n^2)$$

Conclusion

The implementation demonstrates that Algorithm 1 processes weather data much faster than Algorithm 2. As the number of weather records increases, Algorithm 1 remains efficient and scalable, whereas Algorithm 2 requires significantly more computation.

Implementation :

```

1  #include <iostream>
2  #include <vector>
3  #include <algorithm>
4
5  using namespace std;
6
7  // Algorithm 1 : O(n log n)
8  void algorithm1(vector<int> data)
9  {
10     sort(data.begin(), data.end());
11
12     cout << "\nSorted Weather Data:\n";
13     for(int i = 0; i < data.size(); i++)
14     {
15         cout << data[i] << " ";
16     }
17     cout << endl;
18 }
19
20 // Algorithm 2 : O(n^2 log n)
21 void algorithm2(vector<int> data)
22 {
23     sort(data.begin(), data.end());
24
25     long long total = 0;
26     int n = data.size();
27
28     for(int i = 0; i < n; i++)
29     {
30         for(int j = 0; j < n; j++)
31         {
32             total += data[i] * data[j];

```

```

Enter number of weather records: 5
Enter weather values:
5
30
28
31
27
29

Running Algorithm 1 (O(n log n))...

Sorted Weather Data:
5 27 28 30 31

Running Algorithm 2 (O(n^2 log n))...

Total Simulation Value = 14641

=== Code Execution Successful ===

```

Q2. Application: Online Learning Recommendation System

An online learning platform processes user interaction data using the recurrence $T(n) = T(n-1) + n$. Apply the substitution method to expand the recurrence step-by-step and derive the final time complexity. Clearly show intermediate steps and simplify the expression. Analyze how the algorithm performs when the number of users grows rapidly. Interpret the efficiency using asymptotic notation and discuss whether the system is scalable for large educational platforms.

ANSWER:

Algorithm Used

The recommendation system is implemented using recursion based on the recurrence relation

$$T(n) = T(n-1) + n \quad T(n) = T(n-1) + n \quad T(n) = T(n-1) + n$$

The recursive function repeatedly calls itself until the base case is reached.

Input

The program accepts:

- Number of users (**n**)

Output

The program displays:

- Total processing cost after expanding the recurrence.

Recurrence Relation

$$T(n) = T(n-1) + n \quad T(n) = T(n-1) + n \quad T(n) = T(n-1) + n$$

Base Case

$$T(1) = 1 \quad T(1) = 1 \quad T(1) = 1$$

Expansion

$$\begin{aligned} T(n) &= T(n-2) + (n-1) + n & T(n) &= T(n-2) + (n-1) + n & T(n) &= T(n-2) + (n-1) + n \\ T(n) &= T(n-3) + (n-2) + (n-1) + n & T(n) &= T(n-3) + (n-2) + (n-1) + n & T(n) &= T(n-3) + (n-2) + (n-1) + n \\ T(n) &= T(1) + 2 + 3 + \dots + n & T(n) &= T(1) + 2 + 3 + \dots + n & T(n) &= T(1) + 2 + 3 + \dots + n \end{aligned}$$

Using

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Therefore,

$$T(n) = \frac{n(n+1)}{2} \quad T(n) = \frac{n(n+1)}{2} \quad T(n) = \frac{n(n+1)}{2}$$

Ignoring constants,

$$T(n) = O(n^2) \quad \boxed{T(n) = O(n^2)} \quad T(n) = O(n^2)$$

Asymptotic Analysis

Worst Case

$O(n^2)O(n^2)O(n^2)$

Average Case

$\Theta(n^2)\Theta(n^2)\Theta(n^2)$

Best Case

$\Omega(n^2)\Omega(n^2)\Omega(n^2)$

Conclusion

The recursive implementation correctly follows the recurrence relation and demonstrates quadratic growth. The algorithm performs efficiently for small datasets but becomes computationally expensive as the number of users increases, making it less suitable for very large online learning platforms.

Implementation :

```
1 #include <iostream>
2
3 using namespace std;
4
5 long long recommendation(int n)
6 {
7     if(n == 1)
8         return 1;
9
10    return recommendation(n - 1) + n;
11 }
12
13 int main()
14 {
15     int n;
16
17     cout << "Enter number of users: ";
18     cin >> n;
19
20     cout << "\nProcessing..." << endl;
21
22     long long answer = recommendation(n);
23
24     cout << "Total Processing Cost = " << answer << endl;
25
26     return 0;
27 }
```

Enter number of users: 5

Processing...

Total Processing Cost = 15

=== Code Execution Successful ===

Code of Conduct

I am (Rithika K - 192421075), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: (Rithika K)

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____